

28	C	Momentum of particle, $p = \sqrt{2mE}$ Uncertainty in momentum, $\Delta p = 0.010\sqrt{2mE}$ Minimum uncertainty in position, $\Delta x = \frac{h}{\Delta p} = \frac{h}{0.010\sqrt{2mE}} \Rightarrow \Delta x \propto \frac{1}{\sqrt{mE}}$ $\frac{\Delta x_{\text{electron}}}{\Delta x_{\text{baseball}}} = \sqrt{\frac{m_{\text{baseball}} E_{\text{baseball}}}{m_{\text{electron}} E_{\text{electron}}}} = \sqrt{\frac{0.150 \times 100}{(9.11 \times 10^{-31})(1.0 \times 10^6 \times 1.60 \times 10^{-19})}} = 1.01 \times 10^{22}$ Order of magnitude: 10^{22} Note: The mass of the electron is in the data sheet. The mass of the baseball needs to be estimated to the correct order of magnitude.
29	C	An isotope of the parent nuclide will have the same number of protons but different